

Global Energy Storage

Neil Beveridge, Ph.D. +852 2123 2648 neil.beveridge@bernsteinsg.com

Brian Ho, CFA +852 2123 2615 brian.ho@bernsteinsg.com

Katherine Li +852 2123 2694 katherine.li@bernsteinsg.com

Kelvin Yuan, Ph.D., CFA +852 2123 2612 kelvin.yuan@bernsteinsg.com

Battery Weekly 21 September

America

[BYD Takes On The Tesla Semi With Its Own 372-Mile Truck.- insideevs.com](#) BYD unveiled its new ETT 44 electric semi truck at IAA Transportation 2026, positioning it as a competitor to the Tesla Semi in the European market. The truck is powered by a 651 kWh Blade LFP battery, supports 1.5+ MW megawatt charging, and can charge from 20% to 80% in less than 20 minutes, adding approximately 248 miles (400 km) of range. BYD claims a total driving range of 372 miles (600 km) and a maximum gross vehicle weight of 44 tonnes, exceeding the Tesla Semi's European specification. The Blade battery also comes with a 10-year / 1.2 million km warranty, highlighting BYD's push into the growing electric heavy-duty trucking market in Europe.

Asia

[Chinese Firms Upgrade Dubai's Role to Target Market From Transit Hub.- yicaiglobal.com](#) Chinese companies are increasingly treating Dubai and the UAE as end markets for localized operations, rather than simply using Dubai as a trade and logistics hub. Fast-growing investment areas include new energy and batteries, cloud and digital services, advanced mobility, and logistics, with companies increasingly establishing local R&D, assembly, and operating platforms. A survey found 40% of Chinese investors use Dubai primarily to target the UAE and Gulf markets, while 30% use Dubai to expand into Africa and South Asia, 20% prioritize tax and capital-flow advantages, and only 10% primarily target Europe.

[CATL Gets Approval for Battery Deal in Consolidation Push.- BNEF](#) CATL has received regulatory approval to acquire an undisclosed stake in Chongqing Yaoning New Energy Technology, a battery manufacturer backed by Geely, reinforcing China's push to consolidate its increasingly oversupplied battery industry. Beijing has been tightening control over new power and energy-storage battery capacity and encouraging consolidation to curb redundant investment and price competition. The deal could help improve industry utilization and efficiency, although CATL shares fell as much as 6.2% in Shenzhen amid separate market concerns over potential production and earnings cuts.

[SK On to Buy 160 Billion Won of LFP Cathode Materials From L&F.- thelec.net](#) SK On has signed a KRW 160 billion (US\$115 million) agreement with L&F to purchase high-density LFP cathode materials from 2026-2028, with an option to extend for another three years. The materials will support ESS battery production in South Korea and the US, including SK On's Seosan plant, where 3 GWh of its 7 GWh capacity will be converted to LFP ESS batteries for supply from 1H27, as well as converted lines at its Georgia plant. The deal strengthens SK On's non-China LFP supply chain as the company expands its US ESS business.

[Panasonic Energy opens battery development centre in Osaka- electrive.com](#) Panasonic Energy has opened its new Energy Innovation Square battery R&D center in Osaka, bringing materials development, cell design and prototyping under one roof. The facility will employ around 500 people, while Panasonic's combined Osaka battery R&D cluster will have about 800 staff. The company targets 2x development efficiency by FY2031 and expects the integrated setup to cut development time by more than 15%, focusing on next-generation batteries for EVs and AI data centers.

[CATL unveils truck battery with up to 1,000 km range- electrive.com](#) CATL has unveiled its Tectrans II battery system for heavy-duty electric trucks, buses and other commercial vehicles, targeting up to 1,000 km of range and 0-80% charging in 25 minutes using megawatt-level charging. The system offers 170 Wh/kg energy density, around 13% above the industry average, potentially enabling 0.6 tonnes of additional payload, while targeting a 12-year / 1.5 million km lifespan with 70% capacity retention. Its standardized modular design supports different

vehicle architectures and battery swapping, while CATL also signed an agreement with DHL to explore charging, swapping and energy-storage infrastructure for European logistics operations.

[Li Auto replaces CATL cells with its own battery cells- electrive.com](#) Li Auto is gradually replacing CATL batteries with its own 5C fast-charging cells, developed in-house and manufactured by Sunwoda EVB, across models including the L6, L8 and i8. Li Auto recently invested CNY 2.65 billion (US\$390 million) in Sunwoda EVB, increasing its combined stake to 11.17% and becoming the battery maker's second-largest shareholder. The transition will expand from 4Q26, including the upcoming Li 6 for Europe, strengthening Li Auto's control over battery technology, cost, quality and supply.

[CATL Acquires Geely-Backed 30 GWh Battery Plant, Adding Flexibility for EV and ESS Demand.- solarbe.com](#) CATL has received regulatory approval to acquire Chongqing Yaoning New Energy, a Geely-backed 30 GWh battery plant in Chongqing with total planned investment of around RMB 8.5 billion. The acquisition gives CATL a near-complete production base that can serve both EV and energy storage (ESS) demand, while avoiding the time required to build new capacity. CATL's ESS business is growing particularly quickly, with 1H26 ESS revenue up 87.5% YoY to RMB 53.3 billion, shipments reaching about 125 GWh (+81% YoY) and gross margin at 24.0%. The deal also strengthens CATL's southwest China footprint as the company targets more than 1 TWh of total capacity in 2026, although industry-wide ESS overcapacity remains a key risk.

[Namei New Energy Launches 1.5 GWh Sodium-ion Battery Production Line in China.- solarbe.com](#) Namei New Energy Technology (Bazhong) Co., Ltd. (纳美新能源科技) has started production at its 1.5 GWh sodium-ion battery facility in Bazhong, Sichuan, featuring China's first GWh-scale production line for large cylindrical full-tab sodium-ion cells. The project involves around RMB 500 million of investment, with long-term capacity planned to reach 5 GWh and annual output value of RMB 3 billion. The company has already secured RMB 890 million of orders covering energy storage, construction machinery and vehicle starter batteries, supporting the transition of sodium-ion technology from pilot projects toward larger-scale commercialization.

Europe

[BMZ Poland and EVE develop LMFP battery for heavy-duty commercial vehicles- electrive.com](#) BMZ Poland and EVE Energy have formed a strategic partnership to develop the Versora LMFP (lithium manganese iron phosphate) battery platform for electric trucks, buses, construction machinery, agricultural equipment, and marine applications. Compared with conventional LFP batteries, the new LMFP system is expected to be up to 10% lighter, support up to 1 MW charging power, and charge from 10% to 80% in less than 25 minutes, versus around 35 minutes for comparable LFP systems. The batteries are also expected to deliver more than 4,500 cycles while maintaining 80% state of health and retain 85% capacity at -20°C, significantly outperforming traditional LFP batteries in cold-weather conditions. Customer samples are planned for 2027, with series production targeted for mid-2028.

[Porsche produces cathode active materials from recycled raw materials.- electrive.com](#) Porsche, together with German battery recycler Cylib, has produced battery cells whose cathode active materials are made from 100% recycled lithium, nickel, cobalt and manganese recovered from Porsche's own end-of-life batteries. The cells will now undergo testing in Porsche vehicles, with recycled materials potentially available for series-production battery cells from 2028. The project demonstrates a closed-loop battery supply chain that could reduce Porsche's reliance on imported critical raw materials while supporting upcoming EU recycled-content requirements.

[Scania's new battery solution enables 720 km range.- electrive.com](#) Scania is introducing a new Battery Behind Cab solution for electric trucks, allowing additional battery packs to be installed behind the cab alongside existing chassis-mounted batteries. With up to seven battery packs, the maximum configuration can deliver a range of around 720 km, making battery-electric trucks more viable for European long-haul transport. Combined with Scania's new aerodynamic IVD front, energy consumption can be reduced by up to 3%. The solution also supports future MCS megawatt charging, improving the practicality of long-distance electric trucking.

[UK Seeks to Slap Fees on Battery Projects Clogging Up Grid Queue.- BNEF](#) UK energy regulator Ofgem has proposed refundable fees for battery storage projects waiting for grid connections, starting at £3,000/MW and potentially rising to £25,000/MW, to remove speculative projects and accelerate viable developments. The UK currently has around 3x more battery capacity in the connection queue than the system is expected to need by 2035. The maximum fee could equal roughly 25% of capex for short-duration batteries and 7% for medium-duration projects, highlighting the scale of the proposed intervention to ease grid congestion.

[BYD needs 3 assembly plants and 1 battery factory for long-term European expansion, adviser says.- CNEVPOST](#) BYD plans to establish 3 vehicle assembly plants and 1 battery factory in Europe to support its long-term regional growth and comply with EU localization requirements. Its first European plant in Hungary is starting production, while a second assembly location is expected to be selected by end-2026, with Spain and France among the leading candidates. BYD's overseas sales reached 1.16 million vehicles in 8M26, +86% YoY, and management is targeting 1.9–2.0 million overseas sales in 2026 and >2.5 million in 2027, reinforcing the need for additional localized European capacity.

Others

[BME plans battery factory in Egypt with CATL.- electrive.com](https://www.electrive.com) CATL and Egyptian battery manufacturer BME (Battery Manufacturing Egypt) have signed an agreement to establish a battery production facility in Egypt, supporting the country's ambition to build a local battery supply chain. In the first phase, the plant will produce 1 GWh of battery systems annually for heavy-duty commercial vehicles, backed by initial investment of more than EGP 2 billion (US\$39 million). Capacity is planned to expand to 5 GWh in a second phase, adding batteries for passenger vehicles and stationary energy storage systems (ESS). Under the partnership, CATL will provide battery technology, production equipment, and technical support, while BME contributes local manufacturing capabilities. The project aims to achieve 40% local value-added content, reduce battery imports, and position Egypt as a regional battery manufacturing hub for both domestic and export markets.

EXHIBIT 1: Key commodities price performance

	Price 18-Sep	Ccy	Mkt cap \$bn	P/E 26E	EV /EBITDA 26E	%1D	%1W	%1M	%1Y	1Y px chart
Metal prices (US\$/tonne)										
LiCO (spot)	19,606					0.0%	-6%	-10%	91%	
LiOH (spot)	18,115					0.0%	-7%	-10%	74%	
LiCO (contract)	19,750					0.0%	10%	-1%	108%	
LiOH (contract)	18,250					0.0%	-1%	-6%	102%	
Cobalt (spot)	39,350					-0.8%	-5%	-30%	15%	
Nickel (spot)	16,290					0.8%	-1%	-3%	7%	
Component prices										
Cathode NMC811						0.0%	-3%	-9%	24%	
Cathode LFP						0.0%	-4%	-11%	46%	
Precursor NMC811						0.0%	-1%	-9%	5%	
Precursor LFP						0.0%	0%	2%	52%	
Artificial Graphite						0.0%	0%	0%	6%	
Natural Graphite						0.0%	0%	0%	0%	
Electrolyte NMC						0.0%	0%	2%	38%	
Electrolyte LFP						0.0%	0%	3%	59%	
Separator						0.0%	0%	0%	28%	
Spot battery cell cost (US\$/kWh)										
LFP	59					0.0%	-1%	-3%	14%	
NMC532	80					0.0%	-2%	-6%	15%	
NMC622	77					0.0%	-2%	-6%	10%	
NMC811	80					0.0%	-2%	-5%	13%	
Spot battery pack cost (US\$/kWh)										
LFP	72					0.0%	-1%	-2%	-2%	
NMC532	112					0.0%	-1%	-5%	10%	
NMC622	109					0.0%	-1%	-5%	7%	
NMC811	112					0.0%	-1%	-4%	9%	

Source: Bloomberg, Baiinfo, Bernstein analysis and estimates (cell and pack cost)

EXHIBIT 2: Key companies price performance and valuation

	Price 18-Sep	Ccy	Mkt cap \$bn	P/E 26E	EV /EBITDA 26E	%1D	%1W	%1M	%1Y	1Y px chart
Cell makers										
LGES	363,000	KRW	61.4	n.a.	15.0x	-0.5%	1%	4%	2%	
Samsung SDI	539,000	KRW	31.4	79.6x	16.1x	-0.9%	-2%	12%	158%	
CATL	304.1	CNY	214.3	14.7x	10.2x	-0.1%	-8%	-23%	-17%	
CALB	18.5	HKD	4.2	10.3x	2.7x	6.3%	9%	-11%	-33%	
EVE Energy	50.8	CNY	16.5	15.1x	9.6x	5.3%	4%	-10%	-28%	
Gotion	26.0	CNY	7.0	24.0x	5.8x	0.6%	4%	-4%	-42%	
Sunwoda	20.1	CNY	5.5	18.5x	5.9x	-0.8%	9%	10%	-32%	
Farasis	9.1	CNY	1.7	n.a.	n.a.	1.3%	6%	-8%	-57%	
Lithium Miners/Refiners										
Albemarle	115.1	USD	13.6	10.2x	4.9x	5.1%	-2%	-13%	42%	
SQM	71.8	USD	19.5	9.6x	4.6x	4.5%	3%	-4%	67%	
Pilbara Minerals	4.2	AUD	9.6	22.9x	11.6x	-1.1%	-8%	-15%	82%	
Ganfeng Lithium	33.8	HKD	13.1	7.6x	6.9x	1.3%	-3%	-13%	-1%	
Tianqi Lithium	42.6	CNY	10.3	10.0x	3.6x	-0.3%	-1%	-11%	-1%	
Cathode										
LG Chem	266,000	KRW	13.6	n.a.	2.5x	-0.7%	-2%	-1%	-12%	
EcoPro BM	104,200	KRW	7.5	527.8x	49.5x	1.5%	-6%	-1%	-13%	
L&F	110,900	KRW	3.3	n.a.	14.6x	-0.4%	-8%	6%	42%	
Umicore	22.2	EUR	6.3	12.0x	5.3x	0.9%	-1%	-4%	68%	
Ningbo Ronbay	23.8	CNY	2.5	42.6x	11.4x	0.5%	4%	-13%	-14%	
Beijing Easpring	38.8	CNY	3.1	18.9x	12.3x	0.8%	6%	-8%	-30%	
GEM	6.2	CNY	4.7	16.6x	5.2x	0.5%	-1%	-6%	-17%	
Huayou Cobalt	35.5	CNY	10.0	8.7x	3.8x	0.9%	-2%	-15%	-30%	
Dynanonic	41.7	CNY	1.7	13.7x	5.8x	1.5%	6%	-13%	6%	
Anode										
Posco	182,500	KRW	11.7	481.1x	47.2x	0.2%	-2%	15%	31%	
Putailai	22.2	CNY	7.1	14.6x	9.1x	0.2%	4%	-10%	-16%	
Shanshan	11.1	CNY	3.7	n.a.	n.a.	0.6%	-1%	-8%	-27%	
Separator										
Asahi Kasei	1,597	JPY	14.0	14.7x	5.5x	-0.9%	3%	-5%	34%	
SK IE	15,850	KRW	0.9	n.a.	-19.1x	-1.2%	-5%	9%	-45%	
Toray Industries	1,249	JPY	12.0	23.2x	7.3x	-1.9%	1%	-7%	28%	
W-Scope	148	JPY	0.1	n.a.	n.a.	2.8%	-1%	-9%	-42%	
Yunnan Energy	48.3	CNY	7.1	21.3x	8.2x	0.8%	2%	-16%	13%	
Shenzhen Senior	15.6	CNY	3.3	32.1x	10.1x	1.8%	8%	6%	13%	
Sinoma Science	60.1	CNY	15.0	33.6x	14.7x	2.4%	17%	0%	59%	
Electrolyte										
Enchem	20,400	KRW	0.3	n.a.	n.a.	1.0%	-8%	-7%	-75%	
Tinci Minerals	33.1	CNY	10.1	10.5x	7.6x	0.5%	2%	-16%	6%	
Capchem	72.3	CNY	8.1	24.3x	18.2x	-0.2%	1%	6%	52%	
Jiangsu Guotai	7.7	CNY	1.9	9.8x	n.a.	0.5%	2%	-1%	-11%	
Shanshan	11.1	CNY	3.7	n.a.	n.a.	0.6%	-1%	-8%	-27%	
Equipment Maker										
Hirano	1,797	JPY	0.2	17.0x	n.a.	-0.3%	2%	0%	11%	
CKD	5,040	JPY	2.2	27.1x	14.3x	2.3%	0%	-14%	83%	
Wuxi Lead	32.2	CNY	7.9	21.8x	17.5x	1.1%	9%	-9%	-43%	
Shenzhen Yinghe	21.3	CNY	2.0	20.6x	12.4x	1.2%	6%	4%	-39%	
Zhejiang HangKe	20.9	CNY	1.9	23.7x	19.4x	0.7%	4%	-17%	-43%	
Guangdong Lyric	35.9	CNY	0.9	38.8x	15.8x	0.8%	16%	-1%	-50%	
Shenzhen United	21.3	CNY	1.1	21.7x	13.3x	1.1%	7%	3%	-18%	

Source: Bloomberg (all estimates), Bernstein analysis
GLOBAL ENERGY STORAGE

Company Models

[CATL](#) | [LG Energy Solution](#) | [Samsung SDI](#)

[LG Chem](#) | [EcoPro BM](#) | [Posco Future M](#)

[Tianqi Lithium](#)

[Sungrow](#)

Industry Models

[Model: ESS tracker](#)

[Model: Monthly EV Tracker](#)

[Model: Battery Gigafactory Database](#)

[Industry model: Global EV TAM](#)

[Industry model: Global ESS TAM](#)

[Industry model: Battery TAM](#)

[Industry model: Battery supply and demand](#)

Blackbook

[Battery Utopia](#)

[Battery Tech Roadmap 2023: Faster, Higher, Stronger](#)

[Turn Up the Voltage... Investing Across the Battery Value Chain](#)

[Global Energy Storage: Resistance is Futile](#)

Initiation Report

[Global Energy Storage: Initiating Coverage. Lithium Positive \(Tianqi\) and Negative Cathodes \(EcoPro BM, Posco Future M\)](#)

[Sungrow: Solar Eclipse - Initiating at Market-Perform \(PT RMB65\)](#)

Recent Research

[Electric Revolution: Which battery makers are winning the battle for market share](#)

[Global Energy Storage: Data center battery intensity rises as sentiment cools](#)

[Bernstein Energy: Raising battery demand estimates on ESS and CATL target price to HKD770](#)

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[Tianqi Lithium: 1Q26 Best Idea - Energy storage powers lithium upside. Outperform](#)

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[Global Energy Storage: Increasing CATL price target to CNY420 on blockbuster year for ESS](#)

[China EV & Battery Tour Takeaways, 2025 Edition: Steady Evolution](#)

[Global Energy Storage: Can US ESS demand offset EV weakness for Korean battery makers](#)

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BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	17 Sep 2026		TTM	Reported EPS				Reported P/E (x)		
			Closing Price	Price Target	Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
300750.CH (CATL)	O	CNY	304.30	800.00	(39.9)%	CNY	16.14	21.95	28.77	18.9	13.9	10.6
3750.HK (CATL)	O	HKD	507.00	770.00	(22.5)%	CNY	16.14	21.95	28.77	26.8	19.7	15.0
051910.KS (LG Chem)	M	KRW	268,000	312,000	(34.7)%	KRW (13,258.70)	2,693.22	28,743		(20.2)	99.5	9.3
373220.KS (LGES)	M	KRW	365,000	351,000	(20.8)%	KRW (4,580.94)	(2,364.33)	10,225		(79.7)	(154.4)	35.7
006400.KS (SDI)	M	KRW	539,000	520,000	134.1%	KRW (9,933.80)	1,751.66	18,577		(54.3)	307.7	29.0
247540.KS (EcoPro BM)	U	KRW	102,700	140,000	(37.9)%	KRW	403.00	854.00	1,963.00	254.8	120.3	52.3
300274.CH (Sungrow)	O	RMB	84.67	151.00	(64.4)%	RMB	6.55	6.93	7.95	12.9	12.2	10.7
002466.CH (Tianqi Lithium)	O	CNY	42.74	80.00	(24.7)%	CNY	0.28	3.90	7.75	151.5	10.9	5.5
9696.HK (Tianqi Lithium)	O	HKD	30.70	65.00	(45.7)%	CNY	0.28	3.90	7.75	92.9	6.7	3.4
ASIA			1,953.49									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

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